Tertiary Infotech Academy Pte Ltd

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LEARNER GUIDE

For

ISO 9001:2015 Quality Management System

Course Code: C239

Conducted by

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How to Use This Guide

This Learner Guide supports ISO 9001:2015 Quality Management System (C239). It explains the concepts shown visually in the slide deck and contains the detailed procedures for all eight activities. The activity folders contain the matching scenario/checklist PDF and editable Excel workbook.

Important standard-use boundary
This guide paraphrases ISO 9001 concepts for learning. It is not a substitute for the copyrighted standard. Your organisation should use a licensed copy of ISO 9001 for formal requirement interpretation and certification work.

Course Learning Outcomes

- LO1: Identify areas for improvement using an ISO 9001 quality management system and perform process improvement.
- LO2: Apply appropriate quality tools and monitor relevant quality metrics and KPIs.
- LO3: Evaluate processes and assess compliance against quality management system requirements.
- LO4: Follow up the QMS improvement process and document continual improvement according to organisational procedures.

Connected Case Organisation

Name: LionCity Precision Components Pte Ltd (LPC)

Profile: A Singapore SME with 85 employees producing precision aluminium components for commercial equipment.

Customers: Three regional equipment makers account for 72% of revenue; one now requires ISO 9001 certification from key suppliers.

Symptoms: Late delivery rose from 8% to 17%, customer complaints doubled, incoming-material defects increased and procedures vary by shift.

Goal: Build an evidence-based QMS that stabilises order-to-delivery performance and supports certification readiness.

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1. Quality and the ISO 9000 Family

Quality is the degree to which a set of inherent characteristics fulfils requirements. In practice, quality means understanding what must be achieved, designing capable processes, controlling variation, verifying results and learning from evidence.

Term	Practical meaning	Example at LPC
Requirement	A stated, implied or obligatory need	Drawing revision, tolerance, delivery date, certificate content
Conformity	A requirement has been fulfilled	Part dimensions and release evidence meet the approved specification
Nonconformity	A requirement has not been fulfilled	A lot was made to Rev C when the order requires Rev D
Quality management system	Interacting policies, processes, roles, resources and evidence used to direct and control quality	Order review, planning, production, inspection, release, audit and improvement
Effectiveness	The planned activities were performed and the planned results achieved	The drawing-release gate is used and revision errors decline

The ISO 9000 family

Document	Purpose
ISO 9000	Fundamentals and vocabulary for quality management systems.
ISO 9001	Auditable requirements for establishing, implementing, maintaining and continually improving a QMS.
ISO 9004	Guidance for sustained organisational success and maturity beyond minimum requirements.
ISO 19011	Guidance for auditing management systems, including audit principles and programme management.

Certification is a voluntary third-party assurance decision unless a customer, regulator or contract makes it commercially necessary. Certification is granted to an organisation's QMS scope; ISO does not certify organisations directly.

Current status as at 12 August 2026
Teach the published ISO 9001:2015 with Amendment 1:2024 (climate action). ISO states that a revised edition is expected in September 2026; teach only published requirements and plan a gap review after publication.

Seven Quality Management Principles

Principle	What it changes in practice
Customer focus	Understand current and future needs; measure satisfaction; act on changing expectations.
Leadership	Create unity of purpose, direction, accountability and conditions for people to contribute.

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Engagement of people	Build competence, empowerment and recognition across levels and functions.
Process approach	Manage activities as connected processes that produce consistent, predictable results.
Improvement	Make improvement a continuing organisational discipline, not an occasional project.
Evidence-based decision making	Use reliable data and analysis while acknowledging uncertainty and limitations.
Relationship management	Manage relevant relationships, especially suppliers and partners, for sustained value.

Process Approach and PDCA

A process converts inputs into intended outputs through activities, resources and controls. A useful process description identifies purpose, boundaries, ownership, inputs, outputs, customers, requirements, risks, controls, measures and interactions.

PDCA	Management question	Typical evidence
Plan	What result is required, what could affect it and how will the process be controlled?	Objectives, process criteria, risk actions, plans and resources
Do	Are planned controls applied by competent people using suitable resources?	Operational records, approvals, competence and controlled information
Check	What happened, what does the evidence mean and is the system effective?	Measures, analysis, audits, reviews and customer feedback
Act	What must be corrected, improved or standardised?	Corrective actions, improvement plans and updated controls

Risk-based thinking is integrated throughout PDCA. ISO 9001 does not prescribe one risk method. The depth of analysis and control should be proportionate to consequences, uncertainty and the organisation's context.

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2. Understanding ISO 9001 Clauses 4–10

Clauses 4–10 contain the auditable QMS requirements. The summaries below explain intent, practical action and common evidence without reproducing the standard's wording.

Clause area	Paraphrased intent	Examples of objective evidence
4.1 Context	Identify external and internal issues that can affect QMS results; review changes.	Context analysis, strategic risks, review records
4.2 Interested parties	Identify relevant parties and requirements that affect the QMS.	Stakeholder/requirement register, contracts, regulatory register
4.3 Scope	Define boundaries and applicability based on context, parties, products and services.	Controlled scope statement with justified non-applicability
4.4 QMS processes	Define, operate, measure and improve interacting processes.	Process architecture, owners, criteria, measures and records
5.1 Leadership	Demonstrate accountability, customer focus and integration of QMS into business processes.	Decisions, resources, review participation and communication
5.2 Quality policy	Maintain a relevant direction that supports strategy, commitments and objectives.	Approved policy, communication and evidence of understanding
5.3 Roles	Assign and communicate responsibilities and authorities.	Role descriptions, process ownership and decision rights
6.1 Risks/opportunities	Determine and address factors that can affect intended results; evaluate action effectiveness.	Risk actions embedded in processes and subsequent review
6.2 Objectives	Set measurable, relevant objectives and plan who, what, when, resources and evaluation.	KPI dictionary, targets, action plans and trend review
6.3 Changes	Plan QMS changes while considering purpose, consequences, integrity, resources and responsibilities.	Change plan, impact/risk review, approval and verification
7.1 Resources	Provide people, infrastructure, environment, monitoring resources and organisational knowledge.	Capacity plans, maintenance, calibration, knowledge controls
7.2 Competence	Determine competence, act to close gaps and evaluate results.	Competency criteria, training/coaching, observed qualification
7.3 Awareness	Ensure people understand policy, objectives, contribution and consequences.	Interviews, briefings and behaviour at point of work
7.4 Communication	Determine what, when, with whom, how and who communicates.	Communication matrix and traceable internal/external messages
7.5 Documented information	Create, update, control, protect and retain information needed	Approved documents, revision/access control and

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	for effective processes.	retained records
8.1 Operational planning/control	Set process/product criteria, apply controls and retain confidence-building evidence.	Plans, travellers, work controls and release records
8.2 Customer requirements	Communicate, determine, review and manage changes to product/service requirements.	Enquiries, contracts, order review and change confirmation
8.3 Design/development	Plan and control inputs, activities, reviews, verification, validation, outputs and changes when applicable.	Design plan, review evidence, test results and change records
8.4 External providers	Apply risk-based controls to externally provided processes, products and services.	Approval criteria, supplier monitoring and purchase requirements
8.5 Production/service provision	Control execution, identification, property, preservation, post-delivery work and changes.	Work instructions, traceability, process validation and change approval
8.6 Release	Verify acceptance criteria before authorised release.	Inspection/test results and release authority
8.7 Nonconforming outputs	Identify and control outputs that do not conform; retain action evidence.	Segregation/status, concession, correction and re-verification
9.1 Monitoring/analysis/evaluation	Define what and how to measure; analyse results including customer perception.	Metric definitions, valid data, trends, analysis and decisions
9.2 Internal audit	Plan and conduct objective audits of conformity and effective implementation.	Programme, plans, competent independent auditors, evidence and findings
9.3 Management review	Review suitability, adequacy, effectiveness and strategic alignment; make decisions.	Inputs, minutes, improvement/resource/change decisions and actions
10.1 Improvement	Select opportunities that improve products, services, performance and customer satisfaction.	Improvement portfolio, priorities and outcomes
10.2 Nonconformity/corrective action	React, contain/correct, evaluate causes and scope, implement actions and verify effectiveness.	NCR, cause evidence, action plan and effectiveness review
10.3 Continual improvement	Continually improve QMS suitability, adequacy and effectiveness using evaluation outputs.	Trends, review/audit outputs and sustained standardisation

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3. Quality Tools, Measures and Decisions

Choose a tool because it answers a decision question with available data. A polished chart built from inconsistent categories or unreliable samples is not strong evidence.

Tool	Useful for	Quality check
Check sheet	Collect occurrence data consistently at source	Are categories defined? Is collection complete across shifts?
Pareto chart	Prioritise the largest categories	Are counts comparable? Does frequency represent business impact?
Process map/SIPOC	Understand sequence, boundaries and interactions	Does the map show actual work, decisions, loops and evidence?
Fishbone	Structure possible causes	Which hypotheses have objective evidence?
5 Whys	Explore a causal chain	Did the analysis stop at blame or reach a controllable system factor?
Run/control chart	Understand behaviour and variation over time	Is the measure stable, time ordered and interpreted correctly?
Scatter plot	Explore association between two variables	Is there enough paired data, and are confounders considered?
FMEA/risk matrix	Prioritise potential failures and controls	Are scoring definitions consistent and actions proportionate?

Building Useful Quality Objectives and KPIs

- Link the objective to policy, strategy, customer needs and process risk.
- Define the metric precisely: population, numerator, denominator, exclusions, source, frequency and owner.
- Record baseline and target with a time horizon.
- Define the response threshold and decision process before results arrive.
- Use balanced measures so speed, quality, cost and people risks are not optimised in isolation.
- Review data quality and limitations before interpreting movement.

Leading and lagging measures
Lagging measures show results already produced, such as complaints or delivery performance. Leading measures show whether critical controls are operating, such as order-review completion or preventive-maintenance compliance. Use both where they improve decisions.

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Internal Audit and Management Review

An internal audit is a systematic, independent and documented process for obtaining objective evidence and evaluating it against defined criteria. Independence means the auditor is sufficiently free from bias and self-review to reach a fair conclusion; it does not always require an external person.

Finding element	Good practice	Weak practice
Criterion	Identify the applicable requirement or organisational control	Quote a vague expectation
Evidence	Record verifiable facts, sample and traceable references	Record opinion, hearsay or one unverified claim
Statement	Explain the gap between criterion and evidence	Prescribe a solution before cause analysis
Risk	Describe credible consequence and scope	Inflate severity without evidence

Management review is a leadership decision process, not a presentation ceremony. Inputs are analysed to decide improvements, QMS changes and resource needs. Outputs need accountable actions, dates and follow-up.

Nonconformity and Corrective Action

Response	Purpose	Example
Correction	Fix the detected nonconforming output	Rework or replace the affected lot
Containment	Control potentially affected scope and protect customers/processes	Stop shipment; identify WIP and released lots; notify as required
Corrective action	Eliminate a verified cause to prevent recurrence	Redesign drawing-release control and verify sustained reduction

1. Describe the problem with requirement, fact, scope and effect.
2. React: control the issue, correct what can be corrected and address consequences.
3. Determine whether similar issues exist or could occur elsewhere.
4. Identify and verify causes using evidence.
5. Select actions proportionate to consequences and linked to verified causes.
6. Implement actions, update risks/documents/competence and retain evidence.
7. Review effectiveness against pre-defined criteria over a meaningful period.
8. Close, adapt or escalate; preserve learning in standard work and management reporting.

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4. Detailed Activity Procedures

The PowerPoint contains the scenario and decision prompts only. Follow the detailed procedures below and use the files in the corresponding activity folder. Every activity uses the same fictional LPC evidence chain so your decisions carry forward.

Activity A1 — Diagnose Quality Gaps from Evidence

Identify areas for improvement by separating evidence, symptoms and assumptions.

Scenario
LPC's customer complaints doubled in six months. Late deliveries, burr defects and inconsistent certificates are repeatedly mentioned, but managers disagree about the main cause.

Expected output: A completed quality-gap prioritisation matrix and evidence-backed problem statement.

Files: Scenario brief PDF, Quality Gap Workbook (Excel), team discussion

Decision questions

9. Which three issues should LPC prioritise, and what evidence supports each choice?
10. What is a symptom, what is an observed fact and what remains an assumption?
11. How does each issue affect conformity, customer satisfaction, cost or risk?

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Detailed step-by-step procedure

Step 1: Read the scenario and evidence table without proposing solutions yet.

Record the evidence source or workbook cell used. Do not silently convert an assumption into a fact.

Step 2: Classify each item as fact, symptom, possible cause or missing evidence.

Use the supplied template and retain the original scenario evidence. Add missing-data notes where the evidence is insufficient.

Step 3: Score impact, frequency and evidence strength in the workbook.

Show the method, definition or scoring rationale so another person can repeat the analysis.

Step 4: Rank the top three gaps and write one measurable problem statement.

Make the decision explicit: priority, control, owner, target, classification or follow-up as relevant.

Step 5: Peer-review the ranking and record one challenge to your reasoning.

Ask a peer to challenge one important assumption. Record whether you retained or revised the decision and why.

Acceptance test

The top three gaps are traceable to supplied evidence; the problem statement contains a baseline, affected process and measurable effect.

- The output is complete and legible.
- Every conclusion is traceable to scenario evidence or is labelled as an assumption.
- Owners, measures and next decisions are included where applicable.
- The team can explain how the activity supports PDCA and the relevant learning outcome.

Debrief prompts

- What evidence most influenced your decision?
- What uncertainty remains and what additional evidence would reduce it?
- What unintended consequence could your selected action create?
- How will you know that the change is effective and sustained?

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Activity A2 — Map and Improve the Order-to-Delivery Process

Perform process management and improvement using SIPOC, a process map and PDCA.

Scenario
Sales accepts emailed drawings, Planning creates job travellers, Production machines parts, Quality inspects and Logistics ships. Rework and missing revision information frequently surface late.

Expected output: A SIPOC, current-state map, improvement hypothesis and PDCA action plan.

Files: Process Mapping Workbook (Excel), SIPOC checklist PDF

Decision questions

12. Where does value enter and leave the process, and who owns the end-to-end result?
13. Which hand-offs create the greatest risk of wrong revision, delay or rework?
14. What process measures will show whether the future state performs better?

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Detailed step-by-step procedure

Step 1: Define the process purpose, start/end points and customer outcome.

Record the evidence source or workbook cell used. Do not silently convert an assumption into a fact.

Step 2: Complete the SIPOC before drawing detailed activities.

Use the supplied template and retain the original scenario evidence. Add missing-data notes where the evidence is insufficient.

Step 3: Map the current-state sequence, decisions, rework loops and evidence created.

Show the method, definition or scoring rationale so another person can repeat the analysis.

Step 4: Identify one bottleneck and one failure-demand loop using supplied data.

Make the decision explicit: priority, control, owner, target, classification or follow-up as relevant.

Step 5: Design a future-state control and convert it into a PDCA experiment.

Ask a peer to challenge one important assumption. Record whether you retained or revised the decision and why.

Acceptance test

The map has clear boundaries, owners, inputs, outputs, controls and measures; the PDCA experiment has a testable prediction.

- The output is complete and legible.
- Every conclusion is traceable to scenario evidence or is labelled as an assumption.
- Owners, measures and next decisions are included where applicable.
- The team can explain how the activity supports PDCA and the relevant learning outcome.

Debrief prompts

- What evidence most influenced your decision?
- What uncertainty remains and what additional evidence would reduce it?
- What unintended consequence could your selected action create?
- How will you know that the change is effective and sustained?

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Activity A3 — Select and Apply Quality Tools

Choose quality tools that fit the question, data type and customer segment.

Scenario
LPC has 60 recent complaints from three customer segments. The descriptions are inconsistent and teams jump directly to favourite solutions.

Expected output: A coded complaint check sheet, Pareto chart and evidence-led cause hypothesis.

Files: Quality Tools Workbook (Excel), Tool Selection Checklist PDF

Decision questions

15. Which tool answers 'what happens most often' and which explores 'why might it happen'?
16. Do the priorities change by customer segment or product family?
17. Which possible cause needs verification before it can be treated as root cause?

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Detailed step-by-step procedure

Step 1: Code raw complaint descriptions into consistent categories.

Record the evidence source or workbook cell used. Do not silently convert an assumption into a fact.

Step 2: Validate category definitions on a sample with a partner.

Use the supplied template and retain the original scenario evidence. Add missing-data notes where the evidence is insufficient.

Step 3: Build a frequency table and Pareto chart with cumulative percentage.

Show the method, definition or scoring rationale so another person can repeat the analysis.

Step 4: Stratify the vital categories by customer segment.

Make the decision explicit: priority, control, owner, target, classification or follow-up as relevant.

Step 5: Build a fishbone hypothesis and specify evidence needed to test each major branch.

Ask a peer to challenge one important assumption. Record whether you retained or revised the decision and why.

Acceptance test

The Pareto chart reconciles to the complaint total and the selected cause hypothesis is explicitly labelled for verification.

- The output is complete and legible.
- Every conclusion is traceable to scenario evidence or is labelled as an assumption.
- Owners, measures and next decisions are included where applicable.
- The team can explain how the activity supports PDCA and the relevant learning outcome.

Debrief prompts

- What evidence most influenced your decision?
- What uncertainty remains and what additional evidence would reduce it?
- What unintended consequence could your selected action create?
- How will you know that the change is effective and sustained?

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Activity A4 — Set Quality Objectives and KPIs

Set relevant metrics and KPIs for process monitoring and decision-making.

Scenario
LPC's current dashboard lists activity counts but no targets, owners, decision thresholds or links to customer outcomes.

Expected output: A KPI dictionary and dynamic dashboard with targets, formulas, status signals and ownership.

Files: KPI Tracker Workbook (Excel), KPI Design Checklist PDF

Decision questions

18. Which measures are leading, lagging, effectiveness or efficiency indicators?
19. What behaviour could each KPI unintentionally encourage?
20. What decision will management make when a KPI crosses its threshold?

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Detailed step-by-step procedure

Step 1: Rewrite two broad intentions as SMART quality objectives.

Record the evidence source or workbook cell used. Do not silently convert an assumption into a fact.

Step 2: Define each KPI's purpose, formula, unit, data source, owner and frequency.

Use the supplied template and retain the original scenario evidence. Add missing-data notes where the evidence is insufficient.

Step 3: Enter the supplied monthly data and verify formula-driven results.

Show the method, definition or scoring rationale so another person can repeat the analysis.

Step 4: Set target and escalation thresholds with documented rationale.

Make the decision explicit: priority, control, owner, target, classification or follow-up as relevant.

Step 5: Review the dashboard and write the decision triggered by each red status.

Ask a peer to challenge one important assumption. Record whether you retained or revised the decision and why.

Acceptance test

Every KPI has an auditable formula, source, owner, target and defined management response; workbook formulas return no errors.

- The output is complete and legible.
- Every conclusion is traceable to scenario evidence or is labelled as an assumption.
- Owners, measures and next decisions are included where applicable.
- The team can explain how the activity supports PDCA and the relevant learning outcome.

Debrief prompts

- What evidence most influenced your decision?
- What uncertainty remains and what additional evidence would reduce it?
- What unintended consequence could your selected action create?
- How will you know that the change is effective and sustained?

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Activity A5 — Evaluate a Process Against QMS Requirements

Evaluate a process by linking requirements, controls, evidence and results.

Scenario
Supplier approvals exist, but several files are incomplete and incoming defects have increased. Procurement says the process is compliant because a procedure exists.

Expected output: A clause applicability and evidence matrix with conformity judgements and follow-up questions.

Files: Evaluation Matrix Workbook (Excel), Evidence Quality Checklist PDF

Decision questions

21. What evidence shows implementation and effectiveness rather than documentation alone?
22. Which requirements are applicable to this process and why?
23. What additional evidence is needed before reaching a conclusion?

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Detailed step-by-step procedure

Step 1: Define the process purpose, intended results and applicable requirements.

Record the evidence source or workbook cell used. Do not silently convert an assumption into a fact.

Step 2: List the control claimed by LPC and the objective evidence available.

Use the supplied template and retain the original scenario evidence. Add missing-data notes where the evidence is insufficient.

Step 3: Rate evidence for relevance, reliability and sufficiency.

Show the method, definition or scoring rationale so another person can repeat the analysis.

Step 4: Compare actual results with planned results and requirements.

Make the decision explicit: priority, control, owner, target, classification or follow-up as relevant.

Step 5: Record conformity, concern or evidence gap without inventing facts.

Ask a peer to challenge one important assumption. Record whether you retained or revised the decision and why.

Acceptance test

Each judgement cites a requirement, evidence and result; unknowns remain evidence gaps rather than automatic nonconformities.

- The output is complete and legible.
- Every conclusion is traceable to scenario evidence or is labelled as an assumption.
- Owners, measures and next decisions are included where applicable.
- The team can explain how the activity supports PDCA and the relevant learning outcome.

Debrief prompts

- What evidence most influenced your decision?
- What uncertainty remains and what additional evidence would reduce it?
- What unintended consequence could your selected action create?
- How will you know that the change is effective and sustained?

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Activity A6 — Conduct an Internal Audit Simulation

Assess QMS compliance through objective evidence, sampling and impartial reporting.

Scenario
A customer returned parts made to an obsolete drawing revision. LPC needs an impartial internal audit before management review.

Expected output: An audit plan, process-based checklist, evidence log and correctly worded findings.

Files: Internal Audit Workbook (Excel), Audit Checklist PDF, role-play evidence pack

Decision questions

24. What audit trail follows the customer requirement through planning, production, inspection and release?
25. What makes evidence objective and the sample appropriate?
26. Does each finding state requirement, evidence and the gap without prescribing a solution?

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Detailed step-by-step procedure

Step 1: Set audit objectives, scope, criteria, timing and auditor independence.

Record the evidence source or workbook cell used. Do not silently convert an assumption into a fact.

Step 2: Prepare open questions that follow the process and sample records.

Use the supplied template and retain the original scenario evidence. Add missing-data notes where the evidence is insufficient.

Step 3: Interview role-players and record exact evidence, not conclusions alone.

Show the method, definition or scoring rationale so another person can repeat the analysis.

Step 4: Triangulate documents, interviews and observations.

Make the decision explicit: priority, control, owner, target, classification or follow-up as relevant.

Step 5: Classify and write findings; conduct a concise closing meeting.

Ask a peer to challenge one important assumption. Record whether you retained or revised the decision and why.

Acceptance test

The audit trail is reproducible and every nonconformity statement includes the requirement, objective evidence and unmet condition.

- The output is complete and legible.
- Every conclusion is traceable to scenario evidence or is labelled as an assumption.
- Owners, measures and next decisions are included where applicable.
- The team can explain how the activity supports PDCA and the relevant learning outcome.

Debrief prompts

- What evidence most influenced your decision?
- What uncertainty remains and what additional evidence would reduce it?
- What unintended consequence could your selected action create?
- How will you know that the change is effective and sustained?

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Activity A7 — Follow Up a Nonconformity and Corrective Action

Follow up improvement from containment through root cause, action and effectiveness review.

Scenario
Twenty-four parts were produced to revision B after the customer approved revision C. Eight were shipped before the error was detected.

Expected output: A complete NCR/CAPA record with evidence-based root cause and an effectiveness review plan.

Files: CAPA Workbook (Excel), Corrective Action Checklist PDF

Decision questions

27. What correction and containment are needed immediately?
28. What system cause allowed the obsolete revision to reach production?
29. What evidence and time window will demonstrate non-recurrence?

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Detailed step-by-step procedure

Step 1: Record the nonconformity, affected scope, risk and immediate correction.

Record the evidence source or workbook cell used. Do not silently convert an assumption into a fact.

Step 2: Contain suspect product and protect the customer while tracing affected records.

Use the supplied template and retain the original scenario evidence. Add missing-data notes where the evidence is insufficient.

Step 3: Use 5 Whys and cause-and-effect thinking; validate the systemic root cause with evidence.

Show the method, definition or scoring rationale so another person can repeat the analysis.

Step 4: Assign corrective actions to causes with owners, dates and required resources.

Make the decision explicit: priority, control, owner, target, classification or follow-up as relevant.

Step 5: Define and later perform an effectiveness review using trend and audit evidence.

Ask a peer to challenge one important assumption. Record whether you retained or revised the decision and why.

Acceptance test

Actions address verified causes, not only symptoms; the effectiveness criterion is measurable and scheduled after sufficient evidence can accumulate.

- The output is complete and legible.
- Every conclusion is traceable to scenario evidence or is labelled as an assumption.
- Owners, measures and next decisions are included where applicable.
- The team can explain how the activity supports PDCA and the relevant learning outcome.

Debrief prompts

- What evidence most influenced your decision?
- What uncertainty remains and what additional evidence would reduce it?
- What unintended consequence could your selected action create?
- How will you know that the change is effective and sustained?

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Activity A8 — Document the Continual Improvement Report

Document a QMS improvement activity report according to organisational procedures.

Scenario
Three months after LPC launched its improvement programme, management needs evidence of results, unresolved risks and recommended next actions.

Expected output: A structured improvement report with baseline, analysis, decisions, responsibilities, evidence and lessons learned.

Files: Improvement Report Workbook (Excel), Report Quality Checklist PDF

Decision questions

30. What changed relative to baseline and target, and how reliable is the evidence?
31. Which actions are complete, overdue, ineffective or awaiting evidence?
32. What should management decide about resources, changes and standardisation?

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Detailed step-by-step procedure

Step 1: Define the report audience, approval route and required organisational fields.

Record the evidence source or workbook cell used. Do not silently convert an assumption into a fact.

Step 2: Summarise the original problem, baseline, scope and intended result.

Use the supplied template and retain the original scenario evidence. Add missing-data notes where the evidence is insufficient.

Step 3: Present analysis, actions, owners, dates and evidence of implementation.

Show the method, definition or scoring rationale so another person can repeat the analysis.

Step 4: Compare results with targets and state the effectiveness conclusion with limitations.

Make the decision explicit: priority, control, owner, target, classification or follow-up as relevant.

Step 5: Record lessons, residual risks, decisions required and follow-up actions.

Ask a peer to challenge one important assumption. Record whether you retained or revised the decision and why.

Acceptance test

A reviewer can reconstruct the improvement story from evidence and clearly see ownership, status, approval, effectiveness and next review date.

- The output is complete and legible.
- Every conclusion is traceable to scenario evidence or is labelled as an assumption.
- Owners, measures and next decisions are included where applicable.
- The team can explain how the activity supports PDCA and the relevant learning outcome.

Debrief prompts

- What evidence most influenced your decision?
- What uncertainty remains and what additional evidence would reduce it?
- What unintended consequence could your selected action create?
- How will you know that the change is effective and sustained?

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Glossary

Term	Working definition
Audit criteria	Requirements used as a reference against which objective evidence is compared.
Competence	Ability to apply knowledge and skills to achieve intended results.
Conformity	Fulfilment of a requirement.
Corrective action	Action to eliminate the cause of a nonconformity and prevent recurrence.
Documented information	Information required to be controlled and maintained, and the medium containing it.
Interested party	Person or organisation that can affect, be affected by or perceive itself affected by a decision or activity.
Nonconformity	Non-fulfilment of a requirement.
Objective evidence	Data supporting the existence or truth of something.
Process	Set of interrelated activities that transforms inputs into outputs.
Risk	Effect of uncertainty; effects may be positive or negative.
QMS	Quality management system.
Verification	Confirmation through objective evidence that specified requirements have been fulfilled.

References and Further Reading

ISO 9001:2015 standard page: https://www.iso.org/standard/62085.html

ISO 9001 explained: https://www.iso.org/home/insights-news/resources/iso-9001-explained.html

ISO quality management principles: https://www.iso.org/iso/pub100080.pdf

ASQ ISO 9001 overview: https://asq.org/quality-resources/iso-9001

NSF ISO 9001 certification overview: https://www.nsf.org/management-systems/quality-management/iso-9001

Advisera ISO 9001 overview: https://advisera.com/9001academy/what-is-iso-9001/

ISO Library ISO 9001 summary: https://iso-library.com/standard/9001/

ISO 9000 family background: https://en.wikipedia.org/wiki/ISO_9000_family

Accessed for courseware redevelopment on 12 August 2026. Official ISO sources take precedence for the current published status and formal standard information.

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